Is a central break right while the other king is still in the centre?

A coach's moment for a pawn break, on 615 of his quiet middlegame positions and 557 of strong players', refereed by Stockfish

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Abstract

A coach reviewing an adult beginner's rapid games teaches a moment for a pawn break: strike in the centre while the other king has not castled. The app leaves the moment to the engine, because a break that can be played is sound about one time in five in his quiet positions, no more often than a random move. In the positions the structure study scored, his quiet middlegame positions with a central break on, a central break is sound 31% and 30% of the time while the other king is still in the centre, against 14% and 15% once it has left. Measured against a random move in the same positions, the lift is 13.0 points (0.0 to 25.2) on the earlier half of his games and 23.0 (4.7 to 42.4) on the later one. The earlier interval reaches 0, so by the rule fixed before any score was read by the king there is no signal, though both halves point the same way. Strong players' positions do not lean that way: their central breaks are sound 19% of the time with the king in the centre and 18% with it away, and their lift is 5.8 (-3.5 to 15.5). Even with the king in the centre, the central break trails his own move by 28.4 and 26.5 points. The coach's moment may hold at his level, but it is not shown here, and it would not make the break a better choice than his own move: the verdict on a break stays the engine's.

1 The question

A titled player reviewing the rapid games of an adult beginner [1] teaches a moment for a pawn break: strike in the centre while the other king has not castled, because the lines the break opens lead to the king. He adds that players at that level miss the central break because the pawn feels like part of the scenery.

The app teaches each pawn structure's breaks, and two earlier studies shaped what it says about them. A break that can be played is sound only about one time in five in his quiet positions, no more often than a random move, while his own move is sound more than half the time; and the pieces being ready for the break says nothing about whether it is sound, while the app's blunder check does. So the app's lesson tells him to check what a break leaves hanging and leaves the moment to the engine.

The coach's moment is a reading of the board, so it can be tested the same way. In his quiet middlegame positions with a central break on, is the break sound more often when the other king is still in the centre than when it has left, beyond what the king's position does for any move, and is it then as sound as his own move? If the king marks the moment, the app can name a king still in the centre as the time to look at the central break; if the break is then as sound as his own move, as the time to play it. If not, the moment stays the engine's.

2 How it was measured

The positions. The positions the structure study scored, with their scores as they stand. His are his quiet middlegame and endgame positions with a break of his structure on, from his 10-minute games in the app's archive at commit `f4bd63f19b`, in two halves by the games' end time; nothing is fitted, so the halves are two replications. The strong players' are positions from the Lichess games of September 2026 between two players rated 2200 or more, one a game, the first 1,200 scored in source order. A position is quiet when Stockfish's best move is not a capture, a check or a promotion and the mover's win% is between 10 and 90. The referee is Stockfish at the app's grading budget of 250,000 nodes, the hash cleared before each position; a move is sound when it gives up no more than 3 win% against the best.

A position enters this study when it is a middlegame by the app's phase reading, when one of its breaks lands on the d- or e-file (a central break), and when each central break, the move played and the two random moves have exact scores. The other king is *in the centre* when it stands on the d-, e- or f-file, where a king that has not castled stands, and *away* otherwise.

The measures. In each position, C is the share of its central breaks that are sound; A is 1 when his move is sound (for the strong players S, the strong player's move), and R and R_2 the same for two random legal moves that are not breaks, drawn by the structure study. The *lift* is the mean of $C - R$ where the king is in the centre minus its mean where the king is away: R stands for how forgiving a position is, so the lift asks whether the king in the centre helps the central break more than it helps any move. The *strike's gain* is the mean of $C - A$ where the king is in the centre. Intervals come from 2,000 bootstrap draws with seed 20261020, resampling his games within a half (a game's positions of both kinds together), or the strong players' positions: the 95% percentile interval.

3 Counted before the rule

Before the rule was fixed, his positions were counted by the king's file and the breaks' files, with no score read (Table 1). The strong players' positions were being rescored in source order at the time, after a fix to the structure study, and were counted only when the analysis ran. His positions with the king in the centre come from 47 and 39 games, so only a large lift can show on each half.

	His, first half	His, second half	Strong players
King in the centre: positions (games)	116 (47)	83 (39)	129
King away: positions (games)	225 (67)	191 (59)	428
King on its starting square	84	75	114

Table 1: The entering positions: quiet middlegame positions with a central break on, every score the study reads exact. His two halves were counted before the rule was fixed, with no score read; of all the scored positions, his 90 endgames and 204 positions with no central break on stay out, and the strong players' 195 and 448.

4 The rule, fixed before any score was read by the king

The controls come first, and if either fails there is no verdict. (1) His move must be sound more often than R, by an interval above 0, on each half, and the strong player's move on theirs. (2) The lift with R_2 in place of C must not lie above 0 on both halves, nor below 0 on both.

Then, on his positions: the strike holds when the lift lies above 0 on both halves and the strike's gain lies below 0 on neither; the king marks the moment but the break trails his move when the lift lies above 0 on both halves and the gain lies below 0 on a half; the king misleads when the lift lies below 0 on both halves; otherwise there is no signal. There is not enough data when, on a half, the positions with the king in the centre or away come from fewer than 30 games. The strong players' positions are read by the same measures and do not change the verdict.

5 The verdict: no signal

The controls pass (Table 2). His move is sound 37.2 and 38.7 points more often than a random move, and the strong player's 54.8; the null's lift is −3.5 (−13.7 to 6.4) and −0.8 (−11.4 to 10.7), holding 0 on both halves.

The lift is 13.0 points (0.0 to 25.2) on the earlier half and 23.0 (4.7 to 42.4) on the later one. Both are positive, but the earlier interval reaches 0, so by the rule there is no signal. The strong players' positions are silent: their lift is 5.8 points (−3.5 to 15.5), and their central breaks are sound about as often with the king in the centre as with it away (19% and 18%), so the lean in his games does not show in theirs.

The strike's gain answers the second question whatever the first: with the king in the centre, the central break is sound 28.4 and 26.5 points less often than his own move (−38.7 to −18.1 and −43.5 to −9.2), and 58.1 less often than the strong player's.

	His, first half	His, second half	Strong players
Referee: $A - R$ ($S - R$)	37.2 (30.6, 43.3)	38.7 (31.2, 45.7)	54.8 (50.4, 59.2)
Null: $R_2 - R$, centre minus away	−3.5 (−13.7, 6.4)	−0.8 (−11.4, 10.7)	5.0 (−3.7, 13.8)
The lift: $C - R$, centre minus away	13.0 (0.0, 25.2)	23.0 (4.7, 42.4)	5.8 (−3.5, 15.5)
The strike's gain: $C - A$ (S), centre	−28.4 (−38.7, −18.1)	−26.5 (−43.5, −9.2)	−58.1 (−67.9, −48.3)

Table 2: The rule's measures, in points of soundness with 95% intervals from 2,000 bootstrap draws over games. The verdict rests on his halves; the strong players' column is read, not ruled on.

6 What the positions describe

None of what follows was part of the rule (Table 3).

The arms. With the king in the centre a central break is sound 31% and 30% of the time, and 14% and 15% with it away, while a random move is sound 21% and 14% in the first case and 16% and 23% in the second. His own move is sound 59% and 57% with the king in the centre. The central breaks give up 13.4 and 13.5 win% on average with the king in the centre, against 18.6 and 17.9 with it away; his move gives up 4.4 and 5.7.

Other readings. With the king in the centre only when it stands on its starting square, the lift is 11.0 (−5.4 to 28.2) and 19.3 (0.5 to 38.0). With both queens on the board, it is 14.5 (0.8 to 27.1) and 23.3 (2.8 to 42.3). For the breaks that land off the d- and e-files, the same lift is 6.7 (−4.2 to 17.8) and 5.0 (−8.3 to 18.7), holding 0 on both halves. For strong players the stricter reading's lift is 6.7 (−3.4 to 16.9) and the lift with both queens on 4.8 (−6.8 to 15.4), both holding 0, while their flank breaks' lift is 18.0 (7.5 to 28.1), above 0: in strong players' positions it is the flank breaks, not the central ones, that gain on a random move when the king is in the centre.

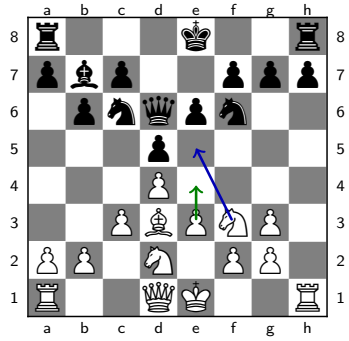
Finding it. He played a central break in 9% and 7% of his positions with the king in the centre, and 7% and 5% with it away. In the positions with both kinds of break on, Maia-2 at his band gives a central break 4.9% and 4.8% on average and a flank break 3.8% and 4.3%: by that reading a central break is no harder for his level to see than a flank break.

	First half, centre	First half, away	Second half, centre	Second half, away	Strong, centre / away
C, the central breaks	31.0 (21.2, 41.3)	13.8 (9.2, 19.2)	30.1 (19.8, 41.8)	15.2 (8.8, 23.0)	19.4 / 18.2
A, his move (S)	59.5 (49.5, 69.2)	52.9 (45.7, 60.2)	56.6 (45.9, 68.9)	59.7 (51.2, 67.8)	77.5 / 68.7
N, Maia-2 at his band	69.0 (59.4, 78.5)	66.2 (59.8, 73.0)	72.3 (62.6, 82.1)	72.8 (65.0, 79.9)	–
U, Maia-2 at the band above	69.0 (59.5, 78.6)	67.6 (60.6, 74.2)	71.1 (61.2, 80.7)	74.3 (66.7, 81.3)	–
R, a random move	20.7 (13.1, 29.5)	16.4 (11.3, 22.0)	14.5 (6.7, 23.2)	22.5 (15.1, 30.1)	12.4 / 17.1
R_2, another	19.8 (12.4, 29.3)	19.1 (13.8, 25.2)	12.0 (5.2, 20.0)	20.9 (13.8, 27.9)	16.3 / 15.9
He played a central break	8.6 (3.6, 14.3)	7.1 (4.0, 11.1)	7.2 (2.6, 12.0)	4.7 (2.0, 7.5)	8.5 / 11.0

Table 3: Described, not tested: the share of each arm that is sound, and how often the move played was a central break, in percent with 95% intervals, where the other king is in the centre and where it is away.

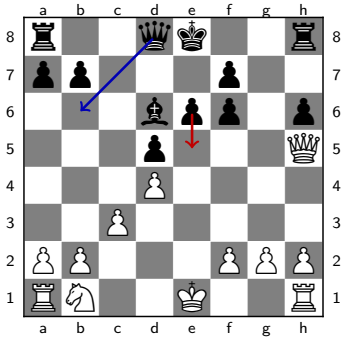
7 Two positions

Two positions of his later half with the other king in the centre (Figure 1): the first where a central break is sound and he played something else, and the first where every central break is unsound and his own move is sound. They illustrate; they test nothing.



FEN `r3k2r/pbp2ppp/1pnqpn2/3p4/3P4/2PBPNP1/PP1N1PP1/R2QK2R w KQkq - 2 10`

(a) He is White, in the London pyramid; the other king is on e8. Stockfish: 10. ♕a4 is best at 55% to win; the central break 10. e4 is sound (55%). He played 10. ♘e5 (blue), not sound at 38%.



FEN `r2qk2r/pp3p2/3bpp1p/3p3Q/3P4/2P5/PP3PPP/RN2K2R b KQkq - 1 13`

(b) He is Black, in the reversed Carlsbad; the other king is on e1. Stockfish: 13. . . ♕b6 is best at 55% to win; the central break 13. . . e5 is not sound (45%). He played 13. . . ♕b6 (blue), sound at 56%.

Figure 1: Two of his positions with the other king in the centre. The break is green where Stockfish finds it sound and red where it does not; win% is his after each move. Stockfish at 250,000 nodes, rerun for this figure.

8 What it means for the app

The app's pawn-break lesson stays as it is: check what a break leaves hanging before playing it, and leave the moment to the engine. A king still in the centre may be a reason to look at the central break, and both halves of his games lean that way, but at this size the lean is not shown, and where it shows most the break is still sound less often than his own move. A test on Lichess games at his band, with many more positions, could settle the lean; it would not change the second answer. Strong players' positions show no lean, so there is nothing to borrow from stronger play either.

9 Caveats

- **Few games.** His positions with the king in the centre come from a few dozen games a half, so only a large lift can show on each half.
- **Not a cause.** A king in the centre comes with an earlier middlegame and more pieces; R stands for how forgiving a position is, not for every difference.
- **A break stands for a plan**, tested as the next move.
- **Mostly the London.** Most of his positions are the London pyramid, whose central breaks are e3-e4 for White and . . . e6-e5 for Black.
- **Mostly blitz.** Most of the strong players' games are blitz; his are rapid.

References

[1] A titled player's review of an adult beginner's rapid games, "Anyone Can Be 1200 In Chess. Watch This." (video, `https://www.youtube.com/watch?v=PcGDehRSqTM`). The lab read a summary of its captions, not the video itself.